

PAPER

Hybrid mental tasks based human computer interface via integration of pronunciation and motor imagery

To cite this article: Jigang Tong *et al* 2022 *J. Neural Eng.* **19** 056048

View the [article online](#) for updates and enhancements.

You may also like

- [Neurophysiological predictors and spectro-spatial discriminative features for enhancing SMR-BCI](#)
Neethu Robinson, Kavitha P Thomas and A P Vinod
- [Feature extraction of four-class motor imagery EEG signals based on functional brain network](#)
Qingsong Ai, Anqi Chen, Kun Chen et al.
- [Riemannian distance based channel selection and feature extraction combining discriminative time-frequency bands and Riemannian tangent space for MI-BCIs](#)
Tingnan Qu, Jing Jin, Ren Xu et al.



PAPER

Hybrid mental tasks based human computer interface via integration of pronunciation and motor imagery

Jigang Tong¹, Xiaoying Wei¹, Enzeng Dong¹ , Zhe Sun^{2,*} , Shengzhi Du^{3,*} and Feng Duan^{4,*}¹ Tianjin Key Laboratory of Control Theory and Applications in Complicated Systems, Tianjin University of Technology, Tianjin 300384, People's Republic of China² Computational Engineering Applications Unit, Head Office for Information Systems and Cybersecurity, RIKEN, Saitama, Japan³ Department of Electrical Engineering, Tshwane University of Technology, Pretoria 0001, South Africa⁴ College of Artificial Intelligence, Nankai University, Tianjin, People's Republic of China

* Authors to whom any correspondence should be addressed.

E-mail: zhe.sun.vk@riken.jp, dushengzhi@gmail.com and duanf@nankai.edu.cn**Keywords:** electroencephalography, pronunciation imagery, Riemannian geometry, hybrid active BCI**Abstract**

Objective. Among the existing active brain–computer interfaces (BCI), the motor imagination (MI) is widely used. To operate the MI BCI effectively, subjects need to carry out trainings on corresponding imagining tasks. Here, we studied how to reduce the discomfort and fatigue of active BCI imaginary tasks and the inability to concentrate on them while improving the accuracy. **Approach.** This paper proposes a hybrid BCI composed of MI and pronunciation imagination (PI). The electroencephalogram signals of ten subjects are recognized by the adaptive Riemannian distance classification and the improved frequency selective filter-bank Common Spatial Pattern recognition. **Main results.** The results show that under the new paradigm with the combination of MI and PI, the recognition accuracy is higher than the MI alone. The highest recognition rate of the proposed hybrid system can reach more than 90%. Furthermore, through the subjects' scoring results of the operation difficulty, it is concluded that the designed hybrid paradigm is more operable than the traditional BCI paradigm. **Significance.** The separable tasks in the active BCI are limited and the accuracy needs to be improved. The new hybrid paradigm proposed by us improves the accuracy and operability of the active BCI system, providing a new possibility for the research direction of the active BCI.

1. Introduction

Brain–computer interfaces (BCI), as a new human–computer interaction mode, provides a convenient and direct communication channel between computers and the brain [1, 2], which has developed rapidly in the fields of medical, military, entertainment and more [3–5]. In the medical field, it can provide an information channel for the disabled or the elderly who have lost the ability to communicate with the world. Nowadays, there are many forms of BCI developed categorized by the bio signals used, such as the electroencephalogram (EEG), cortical EEG, magnetoencephalogram, near-infrared spectroscopy and functional magnetic resonance imaging [6–10], which are then classified as non-invasive and invasive BCI systems [11]. The non-invasive EEG-based BCI systems are not only low cost but also more

convenient and flexible than other methods [12–14]. According to the EEG signals and the methods to generate and capture them, the EEG-based BCI systems are then divided into active, evoked, and passive BCI, such as motor imagination (MI), P300 event-related potential, steady-state visual evoked potential (SSVEP) [15], and the BCI system for emotional EEG signal analysis.

The initial purpose of developing a BCI system is to realize the brain–computer communication and device manipulation without real actions, that is, the 'mind control'. The passive BCI system aims to read users' cognitive states by analysing their brain signal. The classic evoked BCI, such as the BCI based on SSVEP, does not need any training but the long operating time of the dynamic screen will affect users' visual comfort [16], and the low flicker frequency is even easy to cause the seizure of photosensitive

epilepsy [17, 18]. Additionally, this kind of BCIs must rely on external stimulation to induce non-spontaneous EEG signals with corresponding characteristics, which is deviated from the real sense of 'mind control'.

As a typical active BCI system relying on spontaneous EEG, the MI-based BCI (MI-BCI) is different from reactive BCIs. It needs a lot of training due to the spontaneity of the signal [19–23]. Spontaneous EEG also has its unique characteristics, i.e. event-related synchronization/desynchronization (ERS/ERD) [24]. Using these characteristics, MI was first applied by Müller-Putz *et al* in the development of neural prostheses to assist the disabled [25]. Due to the existence of individual differences [26], the user performance of this kind of BCI is uneven. A large number of subjects, in the worst case, are defined as MI-BCI illiteracy [27–30] because the features of their EEG signal are not significantly relevant to the imagination tasks. In addition, the dimensions that can be recognized by simple MI-BCI so far are limited to the movement imagination of hands, feet and tongue [31], which is difficult for subjects without enough training.

In addition to MI-BCI, in recent years, BCI based on silent speech has gradually appeared in the public view. Without producing any actual sounds and facial movements, the EEG signal is collected when subjects imagine phonation [32]. This kind of BCIs has many choices of potential tasks, including the pronunciation imagination (PI) of words, syllables and phonemes. Taking syllable imagination as an example, the EEG can be recognized and classified when subjects silently reading out connected syllables [33, 34]. For word pronunciation, there are also relevant studies with orientation category [35, 36] or 'yes/no' as the object [37–39]. The research of Wang *et al* shows that for imaginative phonation, the effect on the EEG signal can be achieved in the Broca's area and Wernicke's area of the left brain [40].

There are various methods for the identification of active spontaneous EEG signals. The spatial features are commonly extracted from the signal after pre-processing and then classified. Various feature extraction methods based on Common Spatial Pattern (CSP) has been proposed [40, 41].

From the perspective of data enhancement, for example, Pei *et al* proposed a tensor based frequency combination method to conquer the defect of feature selection in FBCSP and construct a feature set, which has low dependence on classifiers [42]. They also proposed a data enhancement method of channel-level recombination, and obtained an effective MI-BCI augmented training set [43].

From the perspective of new classification methods, the EEG classification method based on Riemannian geometry was proposed [44–47], which performs well in EEG data modelling, which becomes the

most advanced and worth mining method in the current BCI research [48].

To improve the operability of active BCIs on the premise of ensuring the system accuracy, we improved the traditional MI-BCI with the following characters. (a) A new active BCI paradigm is proposed with four imagination tasks. Besides the traditional left and right-hand MI, the MI of opening and closing left hand is added as the third imagination task, with the picture of an opened hand as the cue. Based on the subjects' mother tongue (Chinese in this study), the continuous PI of phoneme (Chinese phoneme 'a') is added as the fourth imagination task. (b) In the new paradigm, we replace the traditional arrow-shape cue of the limb MI with a cue of opened hand, which is proved more effective in the experimental results; (c) An improved frequency selective filter bank common space pattern (SFBCSP) algorithm is proposed to classify the EEG signals of the new combined imagination tasks.

Experiments show that the new combinatorial paradigm effectively reduces the discomfort of subjects, therefore, improves the completion rate of tasks. At the same time, the recognition accuracy is improved.

The rest parts of the paper are arranged as follows. Section 2 systematically introduces the experimental methods. Section 3 analyses the experimental results, and the fourth section is an overall discussion and summary.

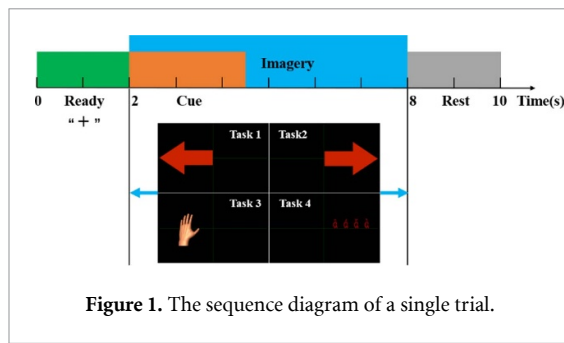
2. Method

2.1. Experimental paradigm

The designed diagram of a single trial is shown in figure 1. The subjects need to complete experiments in an awake state. Four tasks are randomly divided into two sets, according to the cue on the screen: the left arrow and the right arrow form the first set, while the phoneme 'a' of four consecutive tones in Chinese Pinyin and the opened left hand displayed on the screen are in the second set.

In this study, all the subjects are native speakers of Chinese. The Chinese phoneme 'a', as the mother tongue phoneme which is familiar to all subjects, can be easily pronounced. During pronouncing, subjects unfold the mouth shape. Meanwhile, the tongue surface is flat, and the middle of the tongue is slightly concave. As a native speaker of Chinese, the phoneme 'a' is almost the easiest beginning of language education, when used in BCI it would be easier to carry out the corresponding imagination tasks. On the other hand, due to the unique tone fluctuation of Chinese compared with other languages, this selection increased the complexity of the task. The specific description of the tasks are as follows.

When the prompt is a left or right arrow, the subjects imagine the corresponding hand movements.



For the prompt of the opening hand, the subjects open and close the same hand. On the cue of Chinese phonemes appearing on the screen, the subjects imagine making continuous four-tone vocalization of the Chinese phoneme.

The duration of a single trial is 10 s, starting with a cross '+' displayed on the screen lasting for 2 s. At this stage, subjects need to focus their attention. Then the imagination stage starts with a cue appearing for a duration of 2.5 s on the screen. The subjects are supposed to carry out the corresponding imagination task until the 8th second of the trial. After the imagination stage, the screen becomes black for 2 s for a short rest, and the next trial is carried out after then.

As all subjects participated in this experiment for the first time without any BCI experiences, we chose a total of 6 s imagining time including the simple imagining time 3.5 s, and the intercepted time 3.5 s. This is longer than the imagining time of active BCI, but the large amount of repetitive imagining task training in active BCI is avoided. This could improve the comfort of users.

A total of ten subjects participated in the pronunciation-motor-imagination (PMI) experiments and. They were healthy male or female aged 22–26, whose mother tongue is Chinese and all are right-handed. Before the experiments, all ten subjects ensured sufficient rest time without taking mental state related food or chemicals. During the experiments, the subjects keep their hands on the table naturally and stationary with muscles relaxed as much as possible to reduce interference. Each subject conducts 240 trials. To ensure the mental state of the subjects, the trials were divided into four sections separated by 5–10 min intervals to rest.

2.2. Signal acquisition

Figure 2 shows the experimental environment. The experiments were designed in the openVIBE platform [49]. The training task cues are randomly presented on the 24 inch screen according to the pre-designed timing schedule, and the corresponding labels are marked for the EEG signals.

We used 16 channels g. USBamp for EEG signal amplification and acquisition. To be noted that the PMI experiments involve the imagination of both motor and pronunciation tasks, therefore, electrodes



Figure 2. Experimental environment of EEG signal acquisition.

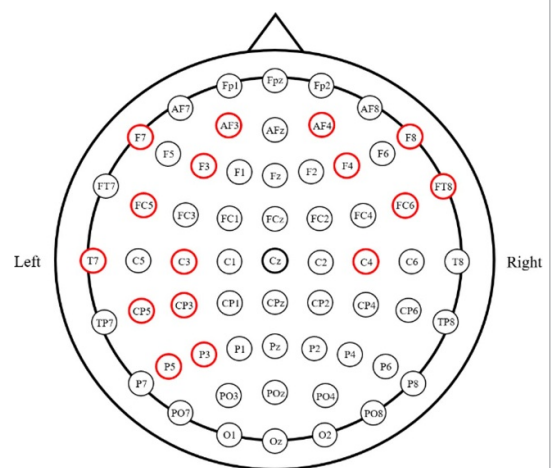


Figure 3. Electrodes selected in the international standard electrode location map.

cover both sensory-motor areas for MI and the areas related to language activities including Broca's area, Wernicke's and visual language centre (inferior parietal horn gyrus, near myopia Centre) of the inferior frontal gyrus. All of language related areas are located in the left brain [50]. A total of 16 channels covering the above positions, according to the international standard 10–20 electrode positioning system, as shown in figure 3 highlighted by red. In addition to the 16 electrodes in red and the reference electrode of the right earlobe, the Cz electrode with bold black circle is grounded. During the experiments, the sampling frequency was set to 256 Hz, and the impedance value of all channels was monitored to be less than 5 kΩ.

2.3. Signal pre-process

During the experiments, subjects need to gaze at the cue on the screen for a long time and repeatedly carry out simple imagination tasks but keep the limbs stationary, there will inevitably be instantaneous tension of body muscles caused by mental fatigue and subconscious blinking. The above situation will lead to the generation of EMG and EOG signals, mixed in the captured EEG signals. These jamming signals exist as artefacts of EEG, which will affect the subsequent signal analysis process. To minimize

the potential interferences in EEG signals, band-pass filtering of relevant frequency bands and the removal of EMG and EOG artefacts are carried out before signal recognition.

Since the brain continuously produces alpha waves (8–13 Hz) and beta waves (14–30 Hz) on awake and concentration states, which are commonly considered relevant to important mental behaviours, therefore, band-pass filtering shall be carried out to remove other background interferences. To avoid losing useful information, we chose to carry out band-pass filtering on the EEG signal with the frequency band of 4–30 Hz.

As the method in this paper was used for off-line BCI, the EEGLAB toolbox was used conveniently to conduct independent component analysis (ICA) on the collected signals both of the control group and experimental group [51], where a variety of signal components are analysed, including EEG, EOG, EMG components and others. We removed the signal components with artefacts (about 30% of total data) manually through the toolbox.

2.4. Signal recognition

After the pre-processing stage of band-pass filtering and ICA, the obtained signals contain important features can be used to classify the tasks done in trials. The CSP algorithm is commonly employed to calculate the features of spatial energy from the pre-processed EEG signals. Machine learning algorithms were then developed for the classification. At this stage, two classification methods are employed for comparison, as shown in figure 4. The first is based on the SFBCSP, an optimized CSP algorithm, which calculates the feature vector with higher discrimination. The second classification method is based on the Riemannian geometry, which extends machine learning to manifold learning without spatial filtering and feature calculation and processing. Different from the traditional classification method, this new classification method does not rely on the parameter adjustment. Both methods compute data covariance. The traditional machine learning method obtains data features directly by processing data covariance, while the Riemannian geometric classifier (RGC) method extends the classification problem to Riemannian manifolds by taking advantage of the characteristics of covariance matrix.

2.4.1. Frequency selective filter-bank common spatial pattern (SFBCSP)

In the traditional machine learning method, we improved the FBCSP, a derivative algorithm of CSP. As shown in figure 5, we searched the EEG of different subjects by searching the frequency band segment, and then calculated the CSP features in the sub-frequency bands.

In the MI-EEG signal classification, the CSP is widely used for spatial energy feature calculation.

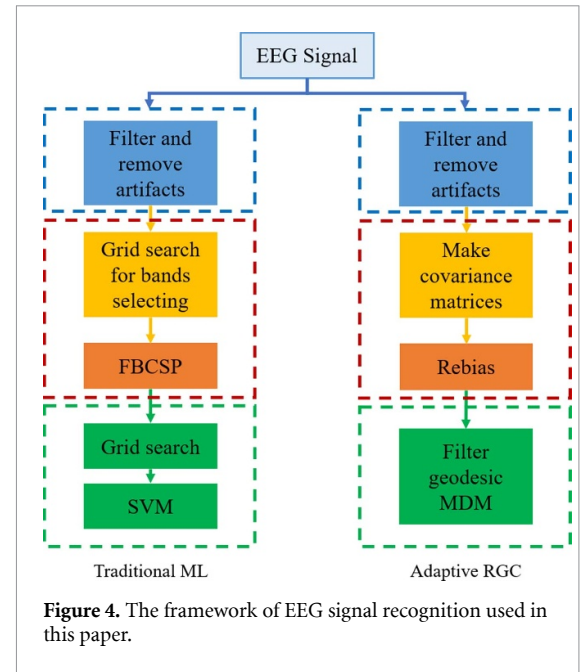


Figure 4. The framework of EEG signal recognition used in this paper.

As shown in the dotted box of figure 5, a group of optimal spatial filters W are found through the diagonalization of the data matrix to project the original signal to a spatial space with a maximized variance between two classes, for a greater discrimination of classes.

The optimal spatial filters W is the key of CSP, which is calculated in few steps. Taking left-right hand MI tasks as an example, with l and r (corresponding to the subscript a and b respectively in figure 5) representing the left hand and the right hand MI related data respectively, the corresponding EEG signals are E_l and E_r . $\bar{C}_l$ and $\bar{C}_r$ represents the mean covariance of the two classes:

$$\bar{C}_l = \frac{E_l E_l^T}{\text{trace}(E_l E_l^T)} \quad (1)$$

$$\bar{C}_r = \frac{E_r E_r^T}{\text{trace}(E_r E_r^T)}. \quad (2)$$

Firstly, the mixed spatial covariance $\bar{C}_m$ is calculated from $\bar{C}_l$ and $\bar{C}_r$, then the whitening matrix A is constructed according to the diagonalized result:

$$\bar{C}_m = \bar{C}_l + \bar{C}_r = U_m \Lambda_m U_m^T. \quad (3)$$

$$A = \Lambda_m^{-\frac{1}{2}} U_m^T. \quad (4)$$

Secondly, as shown in (5) and (6), then the whitening matrix A is used to transform $\bar{C}_l$ and $\bar{C}_r$ to T_l and T_r . In the whitened data T_l and T_r , the correlation between features is minimized but having the same variance, which is helpful for the classification stage. Then, the singular value decomposition is performed on T_l and T_r to obtain the common eigenvector B which is used as the feature for classification. Finally, the spatial filter W is calculated

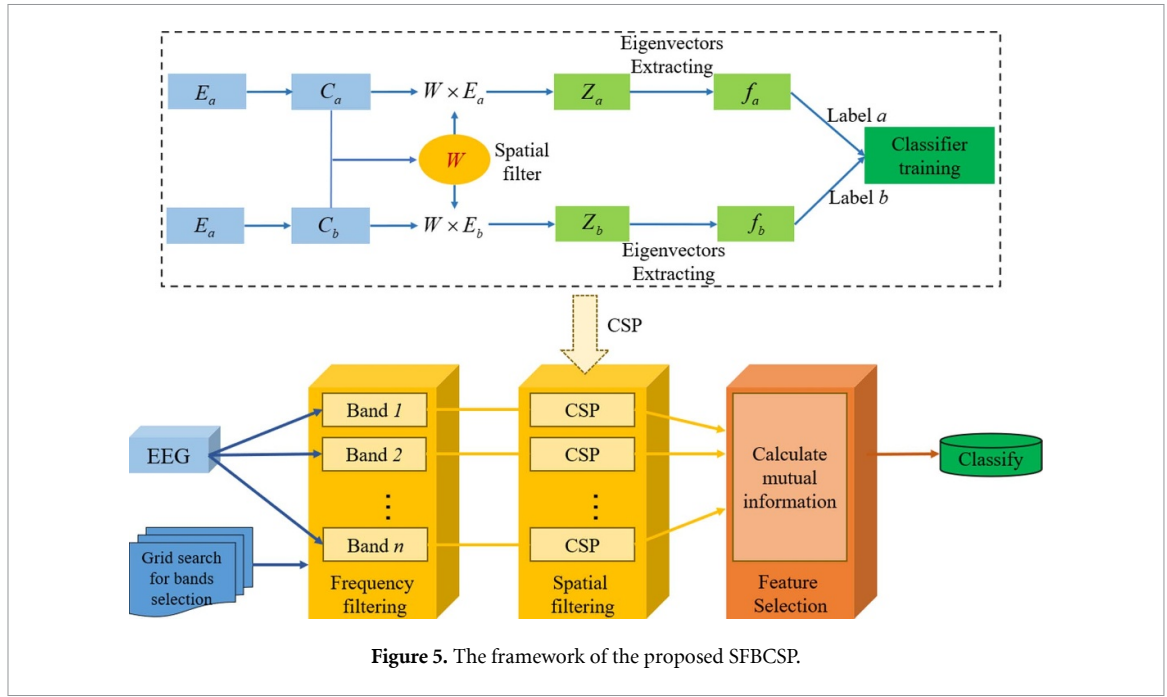


Figure 5. The framework of the proposed SFBCSP.

from equation (7) to project the test data of the two classes to the whitened spatial space, to calculate the corresponding eigenmatrices as features for classification:

$$T_l = AC_l A^T = B\Lambda_l B^T \quad (5)$$

$$T_r = AC_r A^T = B\Lambda_r B^T \quad (6)$$

$$W = (B^T A)^T = A^T B. \quad (7)$$

However, the activities of EEG signals in different rhythms and frequency bands are usually different. Therefore, in the CSP, the signals are not always separable in a frequency band. In the FBCSP, before the calculation of spatial filter W , the EEG signal is divided into n sub-bands by band-pass filters. Then the CSP spatial characteristics are calculated separately for each sub-band. Through the above decomposition and calculation process, the characteristic matrices corresponding to n sub-bands can be obtained. After feature selection from the n feature matrices, the most discrimination features are obtained.

On the other hand, in FBCSP, the signal frequency bands also have different segmentation methods. The bandwidth of EEG frequency bands and sub-bands has impacts on signal recognition and classification as well. Differences exist in EEG signal activity among subjects, so the same frequency band performs quite differently to all subjects in feature calculation. Therefore, if the EEG signal frequency band and the bandwidth of sub-bands are selected properly for each subject, then the overall classification will be better than using the same settings for all subjects.

Besides of the traditional steps, we add the frequency band selection to the FBCSP algorithm. A grid

search method is used to select the range of input EEG signal frequency band and the bandwidth of sub-bands.

Based on the most discrimination features, the classifier is trained. Due to the limited size of the training set, the support vector machine (SVM) is employed, which is commonly used in EEG classification, because of a stable performance and relatively low requirement on the training sample size. However, the SVM needs to adjust multiple parameters, which is considered as its main disadvantage. To this end, we use the 5-fold cross-validation and grid searching methods to adjust the hyper-parameters c and γ to obtain the best classification model.

2.4.2. Adaptive Riemann distance classification

The proposed SFBCSP algorithm in the previous section is based on the correlation covariance among classes. The signal covariance is also used in the RGC, a method for signal classification on Riemannian manifold. Three kinds of RGCs were developed, including minimum distance to the mean (MDM) classifier, Fisher geodesic MDM classifier, and supervised adaptive Fisher geodesic MDM classifier. These RGCs are used for signal classification in the proposed PMI-EEG system.

Here, the PMI-EEG signal is denoted as $E \in \mathbb{R}^{n \times s}$, where n is the number of EEG signal channels and s is the number of sampling points of each channel. The normalized covariance matrix C_E of the signal is shown in equation (8):

$$C_E = \frac{EE^T}{\text{trace}(EE^T)} \quad (8)$$

The C_E is a symmetric positive definite (SPD) matrix [52]. The Riemannian geometry extends EEG signal

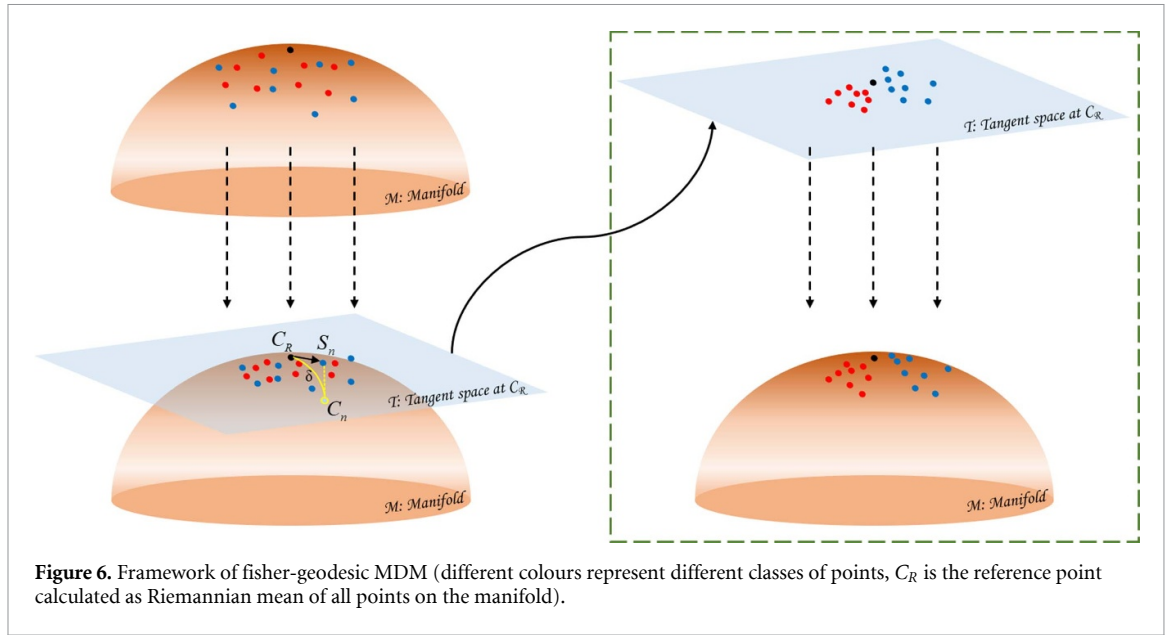


Figure 6. Framework of fisher-geodesic MDM (different colours represent different classes of points, C_R is the reference point calculated as Riemannian mean of all points on the manifold).

data to the manifold for processing, by containing the SPD matrix in the Riemannian geometric domain. Therefore, the Riemannian distance on the manifold where the SPD matrix is located extends the one-dimensional Euclidean geometric distance to a high dimensional space.

As shown by the abstract smooth surface in figure 6, a Riemannian manifold is a smooth manifold differentiable at any point. Every point on the manifold can be expressed as an $n \times n$ SPD matrix forming the set C_n , with a unique corresponding Euclidean tangent space in which all points forming the set S_n . Manifold differentiation can be developed to solve manifold problems linearly.

The geodesic between any two reversible and diagonalizable SPD matrix points C_1 and C_2 on the manifold is calculated by equation (9):

$$\Delta(C_1, C_2, t) = C_1^{\frac{1}{2}} \left(C_1^{-\frac{1}{2}} C_2 C_1^{-\frac{1}{2}} \right)^t C_1^{\frac{1}{2}} \quad t \in [0, 1]. \quad (9)$$

Then the shortest distance between two matrix points can be calculated by equation (10):

$$\begin{aligned} \delta_g(C_1, C_2) &= \int_0^1 \Delta(C_1, C_2, t) dt = \|\log(C_1^{-1} C_2)\|_{\mathcal{F}} \\ &= \left[\sum_{k=1}^n \log^2 \lambda_k \right]^{1/2} \end{aligned} \quad (10)$$

where $\|\cdot\|$ represents the Frobenius norm of a matrix, $\log(\cdot)$ is the logarithm of a matrix, $\lambda_k > 0$ is the eigenvalue of matrix $C_1^{-1} C_2$.

Taking the Riemannian mean of all points on the manifold $C_R \in C_n$ shown in figure 6 as an example, tangent vectors $S_i \in S_n$ are generated at point C_R corresponding to any point $C_i \in C_n$. These tangent

vectors constitute the tangent space T of the manifold M at point C_R , as shown in figure 6. At the same time, tangent vector S_i is corresponding to the geodesic $\Delta_t'(C_R, C_i, 0)$ in (9). The affine invariance of a geodesic, that is, the shortest distance between two matrix points on the manifold, ensures the existence of one-to-one mapping $\text{Log}_{C_R} : C_n \rightarrow S_n$ and reflection $\text{Exp}_{C_R} : S_n \rightarrow C_n$ between the Riemannian manifold and Euclidean tangent space at a point, as shown in equations (11) and (12):

$$\text{Log}_{C_R}(C_i) = C_R^{\frac{1}{2}} \text{Log}(C_R^{-\frac{1}{2}} C_i C_R^{-\frac{1}{2}}) C_R^{\frac{1}{2}} = S_i \quad (11)$$

$$\text{Exp}_{C_R}(S_i) = C_R^{\frac{1}{2}} \text{Exp}(C_R^{-\frac{1}{2}} S_i C_R^{-\frac{1}{2}}) C_R^{\frac{1}{2}} = C_i. \quad (12)$$

With the above equations, we can solve classification problems on the manifold. Through equation (11), the matrix points on the Riemannian manifold are mapped to the Euclidean space, where the conventional methods can be used for data processing and classification. Then the affine invariance in equation (12) maps it back to the Riemannian manifold.

In addition, the Karcher/Fréchet mean is related to the distance defined in equation (10). The Riemannian geometric mean of i SPD matrices $C_1, C_2, \dots, C_i$ is defined as:

$$\bar{C} = \arg \min_{C \in C_n} \sum_{k=1}^i \delta_g^2(C_i, C). \quad (13)$$

In equation (8), we got the covariance matrix C_E of PMI-EEG signal, so the Riemannian geometric mean of the EEG signal can be found using the above process, and then the EEG signals can be classified.

Riemannian distance classification methods based on MDM are considered in this research for PMI-EEG signal classification and recognition.

2.4.2.1. Minimum distance to the mean (MDM)

The MDM algorithm is a basic classification method based on Riemannian distance. The signals collected from dual-task MI-EEG and dual-task PMI-EEG correspond to the left- and right- hands MI, and the opening and closing of the left-hand imagination followed by the imagination of the phoneme pronunciation, respectively. The categories corresponding to the dual-task signals of the MI-EEG and PMI-EEG are denoted as $\lambda_i \in \{1, 2\}$, respectively. After calculating the SPD class covariance matrix C of EEG training data with n channels and s sampling points through equations (8) and (13) is used to calculate the geometric mean C_{1R} and C_{2R} of EEG data corresponding to $\lambda_i = 1$ and $\lambda_i = 2$ on the Riemannian manifold, respectively. These two mean points are also SPD matrix points on the manifold. Using PMI-EEG test data as example, i.e. the tasks in λ_2 , the SPD matrix of a single trial is C_t , then we only need to use equation (10) to calculate the distance $\delta_g(C_t, C_{1R})$ and $\delta_g(C_t, C_{2R})$. The difference of the distances can be obtained in equation (14):

$$x = \delta_g(C_t, C_{1R}) - \delta_g(C_t, C_{2R}). \quad (14)$$

If $x \leq 0$, the class of the trial is predicted as $\lambda_i = 1$, otherwise $\lambda_i = 2$.

However, this method only considers the geometric mean of each class, without considering the distribution of classes. The method described below considers the data distributions of classes, to reduce the impact of this disadvantage.

2.4.2.2. Fisher-geodesic minimum distance to mean (FgMDM)

In this method, the EEG signal will be filtered in the Euclidean tangent space of the reference matrix point C_R , and the Fisher LDA is used to maximize the ratio of across-classes distances to within-classes in the tangent space. Firstly, we need to calculate the Riemannian mean of all matrix points of training data as the reference point C_R by (13). Assuming that there are α matrix points in the two classes, i.e. totally α trials in the two classes in λ_2 . All α matrix points on the Riemannian manifold are projected onto the Euclidean tangent space with C_R as the Riemannian mean, whose tangent space is obtained to calculate $S_i (i \in \{1, \dots, \alpha\})$ by equation (11). Then present S_i upper triangular matrices in the $n(n+1)/2$ dimensional vector form as (15):

$$S_i^v = \text{vec}(S_i). \quad (15)$$

The obtained S_i^v will be used to calculate the most suitable filter F by Fisher LDA method. Thereafter, the filter F is used to the Euclidean tangent space projection S_{ti} of the test data, to obtain S_{ti}^v , as shown in equation (17):

$$F = \text{LDA}(S_i^v) \quad (16)$$

$$S_{ti}^v = F(F^T F)^{-1} F^T S_{ti}. \quad (17)$$

After obtaining the filtered test data in the tangent space, the two groups of filtered data S_i^v and S_{ti}^v can be reflected back into the Riemannian manifold through the process $\text{Exp}_{C_R} : S_n \rightarrow C_n$ as shown in formula (12). The Riemannian mean C_{1R}^v and C_{2R}^v of the matrix points in the labeled training data can be calculated in the manifold before the prediction. To predict the target classes of the test data, we only need to calculate the shortest distance between the filtered test matrix data point to the Riemannian mean points of all classes on the manifold. This step is the same as the last step of MDM.

2.4.2.3. Adaptive fisher-geodesic minimum distance to mean (AFgMDM)

Based on the MDM and FgMDM, the adaptive Riemannian geometric classification is developed for EEG signal classification for PMI-EEG, and compared with the three methods (MDM, FgMDM and the traditional machine learning algorithm). Adaptive algorithms usually add the output results of the classification to the training data and train the classifier iteratively. The Rebias adaptive method is adopted in this paper. In short, the distributions of training and test datasets are adaptively changed by rules, as shown in figure 7. This method aims to minimize the differences of the distributions of the training data and test data. For instance, figure 7 shows the resulting distributions of the two datasets are identical after adaptive changes.

In general, after calculating the reference matrix point C_R^{train} (i.e. Riemannian mean value) for the training data and the updating reference matrix points for the test data, we use the data shifting method proposed by He and Wu [53] to transform the training data and test data distribution as shown in equation (18):

$$C_i' = C_R^{-\frac{1}{2}} C_i C_R^{-\frac{1}{2}} \quad (18)$$

where C_R can be the reference matrix point of training data or test data, and i is the number of trials.

Specifically, For the transformed training data, S_i can be obtain by mapping C_i' from the manifold to the tangent space through equation (11), and S_i^v will be obtained through the same transformation in FgMDM using equation (15). Then the Fisher LDA is used to calculate the filter F for prediction.

For the test data to be evaluated, we have to use equation (19) to obtain the new reference matrix point for the i th test trial:

$$C_{R_i}^{\text{test}} = C_{R_i}^{\text{new}} = \begin{cases} C_R^{\text{train}} & i = 1 \\ C_{i-1} & i = 2 \\ \Delta(C_{R_i}^{\text{new}}, C_i, \frac{1}{i-1}) & i \geq 3 \end{cases} \quad (19)$$

where Δ was defined in equation (9).

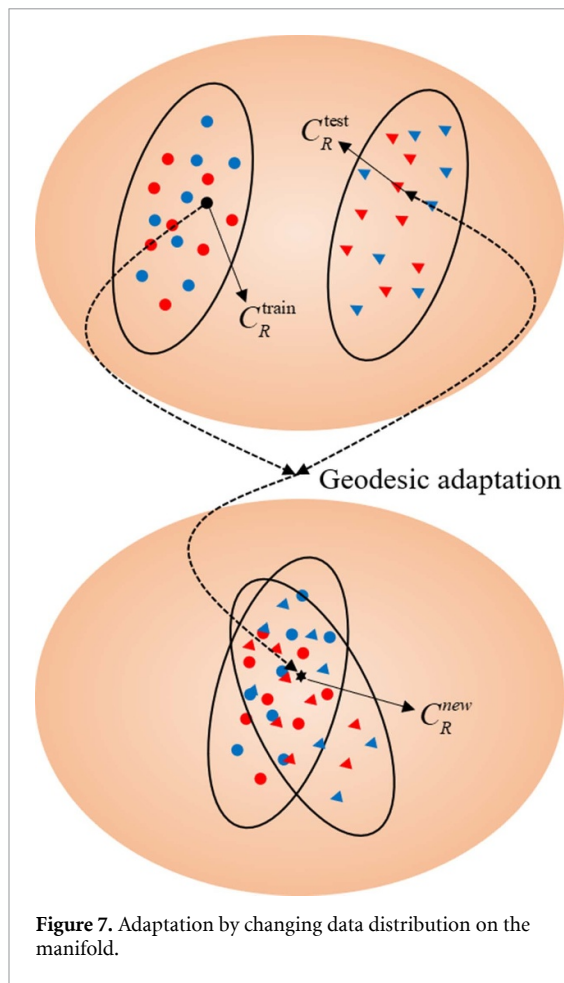


Figure 7. Adaptation by changing data distribution on the manifold.

For regular data updating, after the new reference point of each trial is obtained, the test data distribution is also transformed as the training data through (18), and then mapped to the tangent space by the mapping $\text{Log}_{C_R} : C_n \rightarrow S_n$. Finally, the calculated filter F can be used to obtain S_i^v from equation (17). After reflecting the filtered data S_i^v and S_i^v back to the Riemannian manifold through $\text{Exp}_{C_R} : S_n \rightarrow C_n$, the class of the data is predicted by calculating the shortest distance between the new Riemannian mean points of the classes in training data.

3. Result

3.1. Quantitative results and analysis

The PMI-BCI experiment designed in this research aims to validate the new paradigm and compare with the traditional paradigm of active BCI using MI paradigm alone. Both the performance of signal classification and the comfort of implementing tasks are considered. The adaptive classification method based on Riemannian geometry is applied to the new paradigm.

After the EEG signals of ten subjects were collected for the four predefined tasks, the SFBCSP algorithm and the Riemannian geometry-based classification algorithms (FgMDM, AFgMDM, MDM, and the adaptive MDM with the same adaptive

mechanism as AFgMDM) were employed for comparison across the paradigms and classification methods.

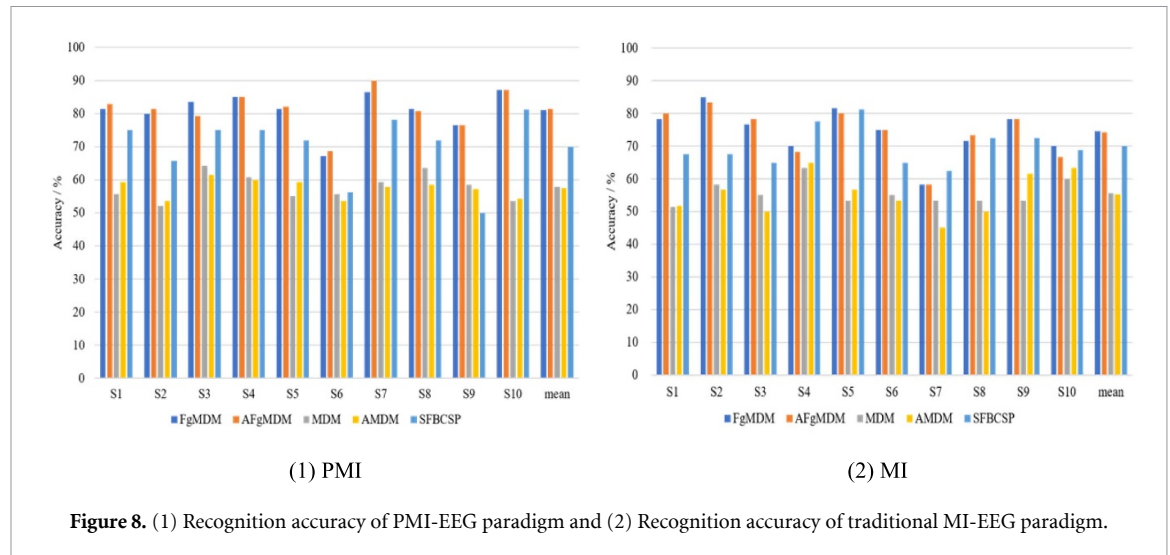
The collected data were divided 1:1 into training and test datasets. The classification results are shown in figure 8, where one finds that the FgMDM and the AFgMDM got higher accuracy in both paradigms, followed by the SFBCSP. When the paradigms are concerned, one finds that the average recognition accuracy of the PMI-EEG paradigm is higher than the traditional MI paradigm. Most subjects improved the accuracy when PMI-EEG is used, except the subject S9. Under the influence of personal differences among subjects and the premise of participating in this kind of experiments for the first time, most subject got about 80% accuracy in FgMDM and AFgMDM. The highest recognition accuracy of 90% is obtained by S7 using AFgMDM in PMI paradigm, which is higher than 60% in the traditional MI paradigm.

Furthermore, three evaluation indicators for each class of data are calculated for comparison: precision, recall rate and F1-score. The precision is the proportion of correct predictions of a class to the total trials predicted as the class. The recall rate is the proportion of correct predictions identified in a class. Obviously, the precision and recall rate are not absolutely positive correlation. The F1-score, harmonic average of the precision and recall rate, can be used as a comprehensive evaluation index. The evaluation values corresponding to the two tasks in the new paradigm (Chinese phoneme with continuous tone and opening-closing hand with a left-hand picture as task cues) and the two tasks in the traditional paradigm (hand MI with left and right arrows as task cues) are shown in tables 1 and 2 respectively.

Comparing tables 1 and 2, one finds that the average performance of the new tasks in PMI paradigm is higher than that in the MI paradigm. The higher index values indicate the stronger separability.

Figure 9 visualizes the box chart comparison of the comprehensive index F1-scores between PMI and MI tasks. One finds that for most subjects, the PMI got higher F1-scores than MI along, which means the proposed PMI is more feasible and considerable than the traditional MI. Most subjects in PMI paradigm got more than F1-scores higher than 0.8, but most in the MI paradigm are lower than 0.8.

On the other hand, the comfort level of the proposed paradigm is compared with the traditional paradigm. It was noted that many people cannot well complete or even understand how to carry out limb MI tasks in the traditional MI paradigm. In fact, a large number of people can only carry out visual MI instead of kinesthetic MI [54], which not only affects the experimental results, but also causes brain fatigue to the subjects. Therefore, after the experiments, a task completion questionnaire is conducted on the subjects to evaluate the experiences of different tasks. Figure 10 shows the average scores for the four tasks,

**Table 1.** Classification performance of the new PMI tasks.

Subject	Phoneme task/Task 4			LH task/Task 3		
	Precision	Recall	F1-score	Precision	Recall	F1-score
S1	0.80	0.87	0.84	0.86	0.79	0.82
S2	0.78	0.87	0.82	0.86	0.76	0.81
S3	0.77	0.83	0.80	0.82	0.76	0.79
S4	0.80	0.93	0.86	0.91	0.77	0.83
S5	0.83	0.81	0.82	0.82	0.83	0.82
S6	0.68	0.72	0.70	0.69	0.65	0.67
S7	0.97	0.83	0.90	0.85	0.97	0.90
S8	0.89	0.70	0.78	0.75	0.91	0.83
S9	0.76	0.75	0.76	0.76	0.77	0.77
S10	0.88	0.85	0.87	0.86	0.89	0.88
Average	0.82	0.82	0.82	0.82	0.81	0.81

Table 2. Classification performance of the traditional MI tasks.

Subject	R task/Task 2			L task/Task 1		
	Precision	Recall	F1-score	Precision	Recall	F1-score
S1	0.85	0.74	0.79	0.76	0.86	0.81
S2	0.82	0.82	0.82	0.84	0.84	0.82
S3	0.73	0.86	0.79	0.85	0.72	0.78
S4	0.78	0.48	0.59	0.64	0.87	0.74
S5	0.83	0.78	0.80	0.78	0.77	0.79
S6	0.71	0.83	0.77	0.80	0.67	0.73
S7	0.58	0.63	0.60	0.59	0.53	0.56
S8	0.70	0.88	0.78	0.80	0.57	0.67
S9	0.78	0.81	0.79	0.79	0.76	0.77
S10	0.76	0.52	0.62	0.62	0.83	0.71
Average	0.75	0.74	0.74	0.75	0.74	0.74

where the PMI tasks got much higher scores than the MI tasks. Most subjects got no difficulty to operate the PMI tasks, but the traditional MI tasks showed varying degrees of difficulty among subjects. Some subjects experienced difficulties to implement the MI tasks and easy to fall asleep. It should be noted that the phoneme imagination got much higher comfort than limb MI, which indicates a new and easier paradigm for the EEG-BCI.

3.2. Energy spectrum analysis

The brain generates different frequency bands of EEG signals in different states, commonly including delta, theta, alpha, beta and gamma waves. The active EEG signals generated by active mental behaviours (such as MI) are different to the passive signal (such as the SSVEP evoked by flashing light stimuli). In the process of soberly and calmly imagining tasks, the alpha and beta bands are mostly significant, which fall in

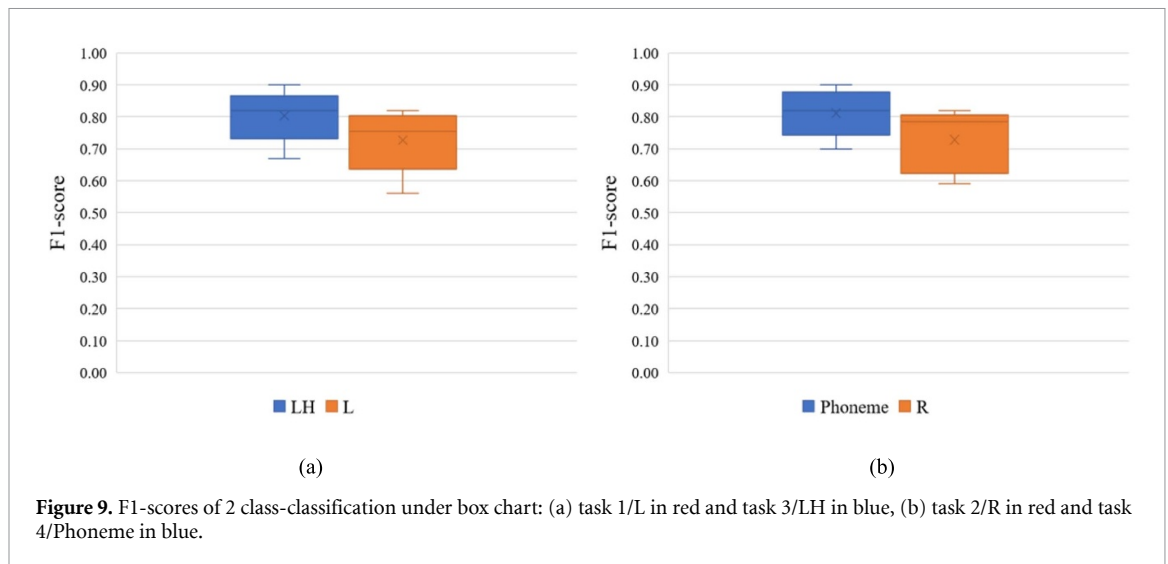


Figure 9. F1-scores of 2 class-classification under box chart: (a) task 1/L in red and task 3/LH in blue, (b) task 2/R in red and task 4/Phoneme in blue.

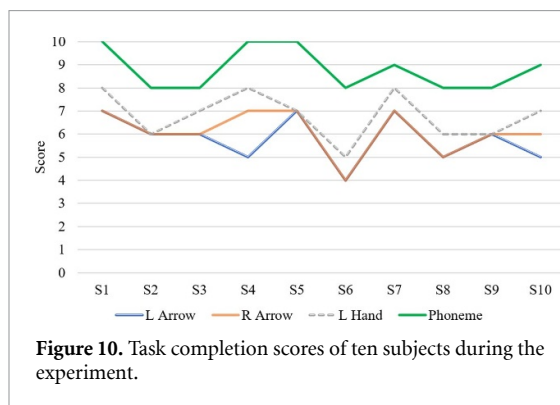


Figure 10. Task completion scores of ten subjects during the experiment.

8–13 Hz and 14–30 Hz bands. The subject S1 (with an average classification performance) was subjected to continuous EEG signal acquisition under two paradigms (left arrow and right arrow vs opening-closing left hand and Chinese phoneme imagination). Signal energy of two EEG rhythms (10 Hz and 20 Hz representing alpha and beta bands respectively) is analysed. The analysis is based on the EEG signals after the filtering and ICA, to minimize the effects from EOG and EMG artifacts. As shown in figure 11, the first two columns of rhythmic energy topographic maps correspond to the 10 Hz and 20 Hz rhythms in the traditional MI paradigm (bilateral hands MI with left and right arrows as task cues). The last two columns correspond to the 10 Hz and 20 Hz rhythms in the proposed new PMI paradigm (pronunciation-MI with opening-closing left hand and the Chinese phoneme PI with four continuous tones as task cues). In figure 11, each energy map corresponds to only one task. For instance, in the first column, the upper energy map is for the task with left arrow as cue and the lower is for the right arrow task. Other columns follow the rule.

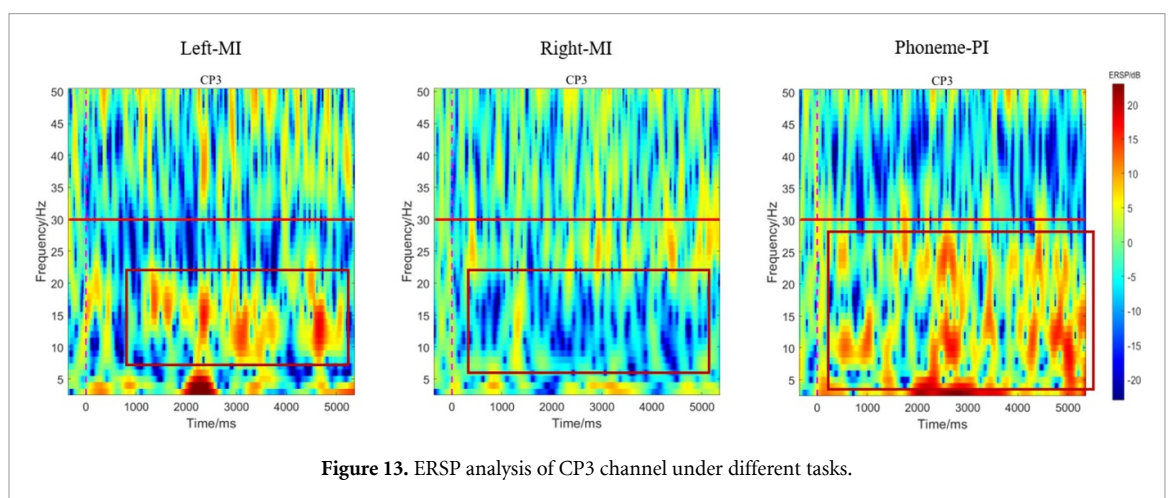
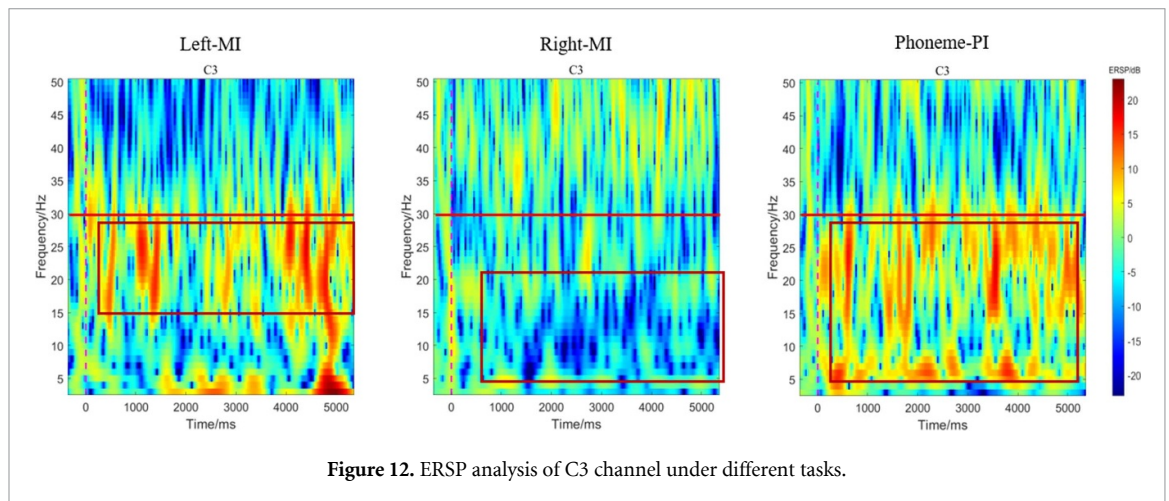
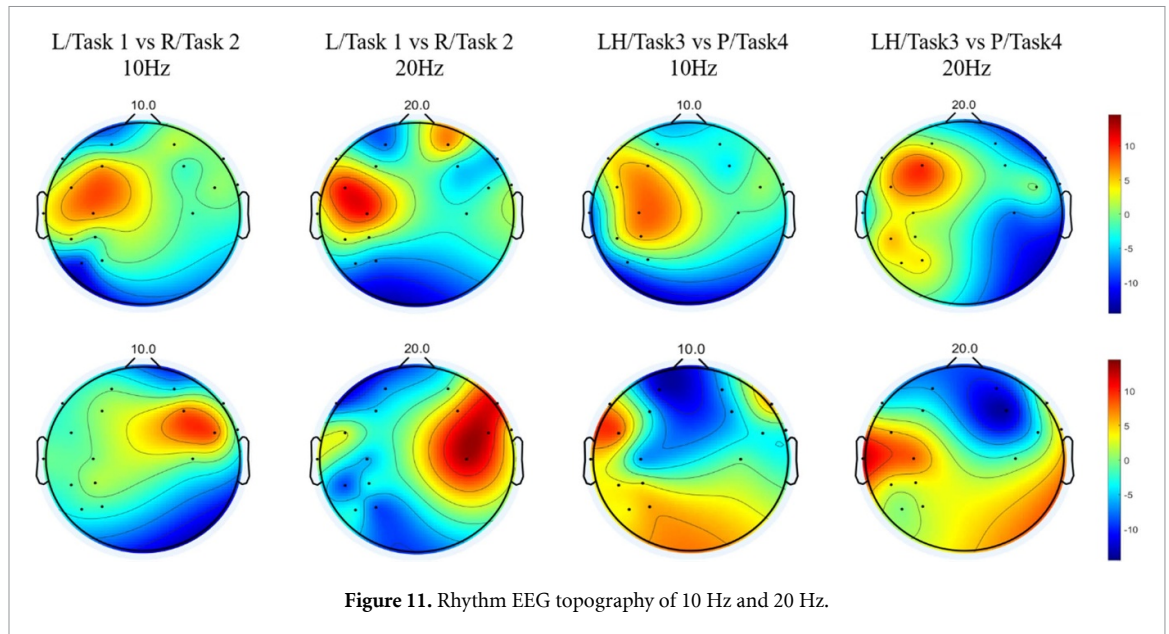
From figure 11, one finds that in the traditional MI paradigm (the first two columns), the activation of brain regions are relatively concentrated. However,

in the PMI paradigm (the last two columns), the activated areas are larger, especially around the language function area of the brain.

In addition, to better observe the impact of paradigm on brain activity, we performed time-frequency domain event-related spectral power (ERSP) analysis on the continuous EEG signals. Taking two important channels among the specific channels as examples. The ERSP of C3 channel is selected since it is in the motor sensory area and commonly used in the EEG analysis of MI tasks. The CP3 channel near the visual language centre is also analyzed. Figures 12 and 13 show the ERSP of the two channels in the three tasks (left- and right- hand MI tasks and phonemic PI) with essential difference in any two, where the concerned regions are highlighted by the rectangle boxes. The energy change of continuous frequency band with the process of mental tasks can be found in figures 12 and 13.

The first column of figures 12 and 13 show the ERSP during left-hand MI tasks, including C3 and CP3 channels (both located on the left brain). The left-hand MI tasks introduced the ERS phenomenon [55] in left brain which is reflected in the red areas of C3 and CP3 channels in the figures. However, because the process of MI mainly depends on the motor sensory area near the central sulcus of the brain, and the position of CP3 channel is closer to the hindbrain than C3 channel, therefore, C3 channel is more affected than CP3 channel in this process.

Similar to the left-hand MI tasks, as shown in the middle column of figures 12 and 13, the ERD phenomenon [55] occurs in both C3 channel and CP3 channel in the left brain during right-hand MI tasks, which is reflected in the dark blue of C3 and CP3 channels in the second column of the two figures. The energy variety in time domain of C3 in alpha and beta frequency bands is slightly larger, reflected in the blue part of CP3 channel diagram that changes little with time.



In the process of phonemic PI task, as shown in the last column in figures 12 and 13, the large area activation phenomenon (the areas of yellow and red colors) is observed, especially in CP3 channel. There is a continuous ERS phenomenon in alpha and beta bands below the 30 Hz dividing line in the figures.

In addition to the time-frequency analysis of the above two important channels, we also show the ERSP analysis results of other selected special channels related to language. As shown in figure 14. From the first line to the fifth line, the ERSP of FT8, T7, CP5, P5 and P3 channels under different tasks are listed in

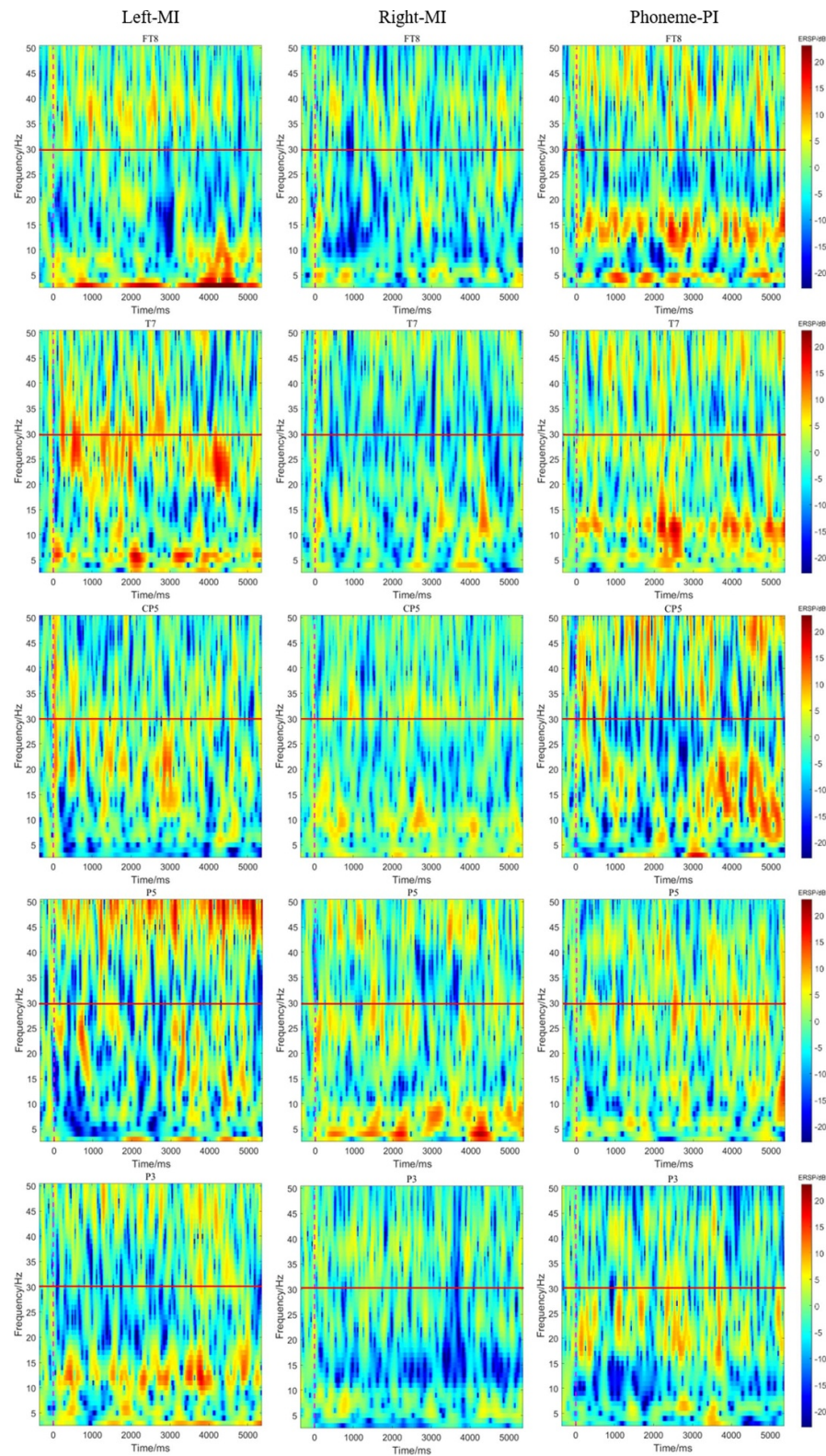


Figure 14. ERSP of analysis of specific channels selected under different tasks.

order. One finds that the FT8 channel near the right auditory center, the T7 channel near Broca's area and the CP5, P5 and P3 channels near Wernicke's area, are closely related to language and have a certain degree of frequency band activation specificity in the process of phonemic PI task, but they are not as obvious as the two channels in figures 12 and 13.

Based on the experiment results, we further demonstrate that the EEG characteristics of the proposed Chinese phoneme imagination with continuous tones overperforms the traditional MI tasks. Combining the two paradigms provides richer features in EEG data, which benefits the classification in BCIs.

4. Discussion

Traditional MI tasks suffering from the low classification accuracy and the problem of MI-BCI illiteracy, the MI-EEG features of some subjects were not clear, and a large portion of users/testers are prone to fatigue or discomfort in repeated imagining tasks. The above problems lead to the fundamental problem of MI BCI system, which is lack of universality and practicability. These problems in active BCI can not be solved by mining from the perspective of algorithm alone. In order to improve the above shortcomings, we had explored from the following two perspectives:

- (a) From the perspective of generating spontaneous EEG mode, or in other words, the perspective of paradigm. In order to improve EEG discrimination, this paper proposed a new paradigm of mixed imagination tasks (limb MI and phonemic PI), combining the Chinese phonemic PI of four continuous tones with the MI tasks. The purpose is to improve the application value of BCI system through the improvement of active BCI paradigm, reduce the difficulty of subjects' operating system to a certain extent, and obtain EEG with greater feature differences. Because PI includes MI in a sense, in the process of speaking, people need the cooperation of multiple brain regions, including motor sensory cortex, visual language center, auditory language center, etc. Therefore, there are multiple brain regions activated in this process, involving frontal lobe, parietal lobe, temporal lobe and parts close to occipital lobe. In addition, during MI, due to the short experimental process, the subjects need to quickly change their thinking and enter the state. The motion prompt is replaced by the corresponding hand instead of the traditional prompt, which improves the efficiency of the subject's state conversion, so as to obtain a more concentrated EEG in a short time. It was proved that the EEG signals of the new paradigm have a higher separability. The time-frequency domain energy analysis showed that the new paradigm can activate a larger area in the brain and a wider EEG frequency band, compared with the traditional MI paradigm. Similarly, other paradigm improvement methods such as integrating virtual reality and augmented reality into BCI have also been proposed to effectively improve BCI performance [56, 57], combining silent reading with writing imagery instead of MI help enhancing MI [58].
- (b) From the perspective of identification method. EEG classification methods include traditional machine learning, adaptive classification, deep learning and matrix and tensor classifier. Riemannian geometry classifier is a new mainstream method of BCI signal recognition based

on matrix and tensor [48]. We developed the adaptive classification method based on Riemannian geometry, the classification problem is extended to Riemannian manifold through the properties of SPD matrix, and the rebias adaptive method is introduced to compare the classification effects of MDM and FgMDM classification methods. The results showed that this method is more applicable than the classical machine learning methods based on CSP, which is reflected in the following two aspects. The classification method based on Riemannian geometry omits the complex parameter adjustment process, and the classification effect is more significant than the traditional method, and the adaptive classification method based on fgmdm performs better than the adaptive classification method based on MDM.

5. Limitation and future works

Firstly, the new active BCI paradigm proposed in this paper aims to realize a system with higher performance, stronger operability and universality than the motion imagination BCI system, but the number of instructions in practical application is also one of the important indicators of the system. The research of this paper is in the initial stage and needs further instruction expansion at the application level. At the same time, our proposed method reduces the impact of poor EEG signal recognition from the paradigm level, no attempt was made to identify signals across subjects from the level of data, so as to weaken the impact of individual differences.

Secondly, in this paper, we empirically selected 16 asymmetric electrodes for EEG acquisition based on brain functional zones in addition to the reference electrodes, which has a certain impact on the overall brain mapping analysis. Corresponding to other analysis results, we compared the brain mapping analysis and take it as a verification.

Therefore, in the next experiment, we will firstly try to add different types of mental tasks from the perspective of mixed imagination, ensure a certain system accuracy while increasing the instruction dimension, and make the subjects have no discomfort during the experiment as far as possible. In addition, individual differences have a great impact on active BCI. EEG signals of different subjects can be significantly different. Improving system universality from the perspective of cross-subjects is also the direction we will try. Secondly, in order to carry out EEG analysis more effectively, we will conduct EEG signal acquisition experiments with complete coverage of brain electrodes in follow-up research, so as to carry out more detailed channel analysis and selection. How to reduce the imagining time will also be considered to increase the operation efficiency of BCI.

6. Conclusion

Active BCI is an important development direction in BCI interface system. In this paper, a new hybrid imagination paradigm based on the combination of continuous tone Chinese phonemic PI and MI is proposed. At the same time, the adaptive classification method based on Riemannian geometry is applied to the process of signal recognition, which verifies the advantages of the new active BCI compared with the traditional paradigm, stable and high signal recognition effect can be obtained under the condition of ensuring that the untrained subjects have no discomfort in the experimental operation.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgments

This work was partially supported by the Natural Science Foundation of Tianjin (No. 18JCY-BJC87700), and the South African National Research Foundation Incentive (No. 81705). National Key Research and Development Program of China (No. 2017YFE0129700), the National Natural Science Foundation of China (Key Program) (No. 11932013), the Tianjin Natural Science Foundation for Distinguished Young Scholars (No. 18JCJQJC46100).

Ethical statement

All procedures performed in the study involving human participants were in accordance with the ethical standards of the Institutional Review Board at Tianjin University of Technology and with the 1964 Helsinki declaration and its later amendments. Informed consent was obtained from all individual participants included in the study. And there is no conflict of interest exists.

ORCID iDs

Enzeng Dong  <https://orcid.org/0000-0001-5142-5584>

Zhe Sun  <https://orcid.org/0000-0002-6531-0769>

References

- [1] Gao S, Wang Y, Gao X and Hong B 2014 Visual and auditory brain-computer interfaces *IEEE Trans. Biomed. Eng.* **61** 1436–47
- [2] Vaughan T, Heetderks W, Trejo L, Rymer W and Wolpaw J 2003 Brain-computer interface technology: a review of the second international meeting *IEEE Trans. Neural Syst. Rehabil. Eng.* **11** 94
- [3] Birbaumer N, Chaudhary U and Ramos-Murguialday A 2016 Brain-computer interfaces for communication and rehabilitation *Nat. Rev. Neurol.* **12** 513–25
- [4] Edelman B J, Meng J, Suma D, Zurn C, Nagarajan E, Baxter B S, Cline C C and He B 2019 Noninvasive neuroimaging enhances continuous neural tracking for robotic device control *Sci. Robot.* **4** eaaw6844
- [5] Scherer R, Muller G R, Neuper C, Graitmann B and Pfurtscheller G 2004 Brain-computer interfaces (BCI)-an asynchronously controlled EEG-based virtual keyboard: improvement of the spelling rate *IEEE. Trans. Biomed. Eng.* **51** 979–84
- [6] Xu J, Mitra S, Hoof C V, Yazicioglu R F and Makinwa K A A 2017 Active electrodes for wearable EEG acquisition: review and electronics design methodology *IEEE Rev. Biomed. Eng.* **10** 187–98
- [7] Zhang J, Sudre G, Li X, Wang W, Weber D J and Bagic A 2011 Clustering linear discriminant analysis for meg-based brain computer interfaces *IEEE Trans. Neural Syst. Rehabil. Eng.* **19** 221–31
- [8] Schalk G and Leuthardt E C 2012 Brain-computer interfaces using electrocorticographic signals *IEEE Rev. Biomed. Eng.* **4** 140–54
- [9] Sitaram R, Weiskopf N, Caria A, Veit R, Erb M and Birbaumer N 2008 fMRI brain-computer interfaces *IEEE Signal Process. Mag.* **25** 95–106
- [10] Watanabe K, Tanaka H, Takahashi K, Niimura Y and Kurihara Y 2016 NIRS-based language learning BCI system *IEEE Sens. J.* **16** 2726–34
- [11] Lebedev M and Nicolelis M 2006 Brain-machine interfaces: past, present and future *Trends Neurosci.* **29** 536–46
- [12] Collinger J L et al 2014 Collaborative approach in the development of high-performance brain-computer interfaces for a neuroprosthetic arm: translation from animal models to human control *Clin. Transl. Sci.* **7** 52–59
- [13] Townsend G and Platsko V 2016 Pushing the P300-based brain-computer interface beyond 100 bpm: extending performance guided constraints into the temporal domain *J. Neural Eng.* **13** 026024
- [14] Zhang H, Zhao X, Wu Z, Sun B and Li T 2021 Motor imagery recognition with automatic EEG channel selection and deep learning *J. Neural Eng.* **18** 016004
- [15] Abiri R, Borhani S, Sellers E W, Jiang Y and Zhao X 2019 A comprehensive review of EEG-based brain-computer interface paradigms *J. Neural Eng.* **16** 1–21
- [16] Xu M, Han J, Wang Y and Ming D 2017 Optimizing visual comfort and classification accuracy for a hybrid P300-SSVEP brain-computer interface 2017 8th Int. IEEE/EMBS Conf. on Neural Engineering (NER) pp 363–6
- [17] Lin F, Zao J, Tu K, Wang Y, Huang Y, Chuang C, Kuo H, Chien Y, Chou C and Jung T 2012 SNR analysis of high-frequency steady-state visual evoked potentials from the foveal and extrafoveal regions of human retina *Conf. Proc. IEEE Engineering Medicine Biology Society* pp 1810–4
- [18] Wilkins A, Veitch J and Lehman B 2010 LED lighting flicker and potential health concerns: IEEE standard PAR1789 update 2010 *IEEE Energy Conversion Congress and Exposition, ECCE 2010—Proc.* pp 171–8
- [19] Matthias W, Erika K, Manuel N, Christa N and Guilherme W 2013 Control beliefs can predict the ability to up-regulate sensorimotor rhythm during neurofeedback training *Front. Hum. Neurosci.* **7** 478
- [20] Badia S B, Morgade A G, Samaha H and Verschure P F M J 2013 Using a hybrid brain computer interface and virtual reality system to monitor and promote cortical reorganization through motor activity and motor imagery training *IEEE Trans. Neural Syst. Rehabil. Eng.* **21** 174–81
- [21] Suwannarat A, Pan-ngum S and Israsena P 2018 Comparison of EEG measurement of upper limb movement in motor imagery training system *Biomed. Eng. Online* **17** 103
- [22] Zapala D, Francuz P, Zapala E, Kopsi N, Wierzgala P, Augustynowicz P, Majkowski A and Kolodziej M 2018 The impact of different visual feedbacks in user training on motor imagery control in BCI *Appl. Psychophysiol. Biofeedback* **43** 23–35

- [23] Choi K 2013 Electroencephalography (EEG)-based neurofeedback training for brain-computer interface (BCI) *Exp. Brain Res.* **231** 351–65
- [24] Pfurtscheller G, Brunner C, Schlögl A and Lopes da Silva F H 2016 Mu rhythm (de)synchronization and eeg single-trial classification of different motor imagery tasks *Neuroimage* **31** 153–9
- [25] Müller-Putz G R, Scherer R, Pfurtscheller G and Rupp R 2005 EEG-based neuroprosthesis control: a step towards clinical practice *Neurosci. Lett.* **382** 169–74
- [26] Graimann B, Pfurtscheller G, Allison B and Neuper C 2010 *Brain-Computer Interfaces* (Berlin:Springer) (<https://doi.org/10.1007/978-3-642-02091-9>)
- [27] Lee M, Kwon O, Kim Y, Kim H, Lee Y, Williamson J, Fazli S and Lee S 2019 EEG dataset and OpenBMI toolbox for three BCI paradigms: an investigation into BCI illiteracy *GigaScience* **8** 1–15
- [28] Ahn M, Cho H, Ahn S and Jun S 2013 High theta and low alpha powers may be indicative of BCI-illiteracy in motor imagery *PLoS One* **8** e80886
- [29] Vidaurre C and Blankertz B 2009 Towards a cure for BCI illiteracy *Brain Topogr.* **23** 194–8
- [30] Allison B, Luth T, Valbuena D, Teymourian A, Volosyak I and Graser A 2010 BCI demographics: how many (and what kinds of) people can use an SSVEP BCI *IEEE Trans. Neural Syst. Rehabil. Eng.* **18** 107–16
- [31] Gouy-pailler C, Congedo M, Brunner C, Jutten C and Pfurtscheller G 2010 Nonstationary brain source separation for multi-class motor imagery *IEEE Trans. Biomed. Eng.* **57** 469–78
- [32] Torres-García A A, Reyes-García C A, Villaseñor-Pineda L and García-Aguilar G 2016 Implementing a fuzzy inference system in a multi-objective EEG channel selection model for imagined speech classification *Expert Syst. Appl.* **59** 1–12
- [33] D’Zmura M, Deng S, Lappas T, Thorpe S and Srinivasan R 2009 Toward EEG sensing of imagined speech *Int. Conf. on Human-Computer Interaction* pp 40–48
- [34] Zhao S and Rudzicz F 2015 Classifying phonological categories in imagined and articulated speech 2015-August *Int. Conf. on Acoustics Speech and Signal Processing* pp 992–6
- [35] Qureshi M, Min B, Park H, Cho D, Choi W and Lee B 2017 Multiclass classification of word imagination speech with hybrid connectivity features *IEEE Trans. Biomed. Eng.* **65** 2168–77
- [36] González-Castaneda E F, Torres-García A A, Reyes-García C A and Villaseñor-Pineda L 2016 Sonification and textification: proposing methods for classifying unspoken words from EEG signals *Biomed. Signal Process. Control* **37** 82–91
- [37] Sereshkeh A R, Yousefi R, Wong A T, Rudzicz F and Chau T 2019 Development of a ternary hybrid fNIRS-EEG brain-computer interface based on imagined speech *Brain-Comput. Interfaces* **4** 128–40
- [38] Sereshkeh A R, Trott R, Bricout A and Chau T 2017 EEG classification of covert speech using regularized neural networks *IEEE/ACM Trans. Audio Speech Lang. Process.* **25** 2292–300
- [39] Hashim N, Ali A and Mohd-Isa W N 2018 Word-based classification of imagined speech using EEG *Comput. Sci. Technol.* **488** 195–204
- [40] Wang L, Zhang X, Zhong X and Zhang Y 2013 Analysis and classification of speech imagery EEG for BCI *Biomed. Signal Process. Control* **8** 901–8
- [41] DaSalla C S, Kambara H, Sato M and Koike Y 2009 Single-trial classification of vowel speech imagery using common spatial patterns *Neural Netw.* **22** 1334–9
- [42] Pei Y, Luo Z, Zhao H, Xu D, Li W, Yan Y, Yan H, Xie L, Xu M and Yin E 2022 A tensor-based frequency features combination method for brain-computer interfaces *IEEE Trans. Neural Syst. Rehabil. Eng.* **30** 465–75
- [43] Pei Y, Luo Z, Yan Y, Yan H, Jiang J, Li W, Xie L and Yin E 2021 Data augmentation: using channel-level recombination to improve classification performance for motor imagery EEG *Front. Hum. Neurosci.* **15** 113
- [44] Guan S, Zhao K and Yang S 2019 Motor imagery EEG classification based on decision tree framework and riemannian geometry *Comput. Intell. Neurosci.* **2019** 1–13
- [45] Gaur P, Pachori R B, Wang H and Prasad G 2018 A multi-class EEG-based BCI classification using multivariate empirical mode decomposition based filtering and Riemannian geometry *Expert Syst. Appl.* **95** 201–11
- [46] Gaur P, Mccreadie K, Pachori R B, Wang H and Prasad G 2019 Tangent space features-based transfer learning classification model for two-class motor imagery brain-computer interface *Int. J. Neural Syst.* **29** 1950025
- [47] Wang T, Du S and Dong F 2021 A novel method to reduce the motor imagery BCI illiteracy *Med. Biol. Eng. Comput.* **59** 2205–17
- [48] Lotte F, Bougrain L, Cichocki A, Clerc M, Congedo M, Rakotomamonjy A and Yger F 2018 A review of classification algorithms for EEG-based brain-computer interfaces: a 10-year update *J. Neural Eng.* **15** 031005
- [49] Inria: OpenVibe Software for brain computer interfaces and real time 2015 (available at: <http://OpenVibe.inria.fr>) (Accessed 17 November 2017)
- [50] Bocquelet F, Hueber T, Girin L, Chabardès S and Yvert B 2016 Key considerations in designing a speech brain-computer interface *J. Physiol.* **110** 392–401
- [51] Delorme A and Makeig S 2004 EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis *J. Neurosci. Methods* **134** 9–21
- [52] Jayasumana S, Hartley R, Salzmann M, Li H and Harandi M 2013 Kernel methods on the Riemannian manifold of symmetric positive definite matrices, 2013 *IEEE Conf. on Computer Vision and Pattern Recognition* pp 73–80
- [53] He H and Wu D 2020 Transfer learning for brain-computer interfaces: an Euclidean space data alignment approach *IEEE Trans. Biomed. Eng.* **67** 399–410
- [54] Chholak P, Niso G, Maksimenko V A, Kurkin S A, Frolov F S, Pitsik E N, Hramov A E and Pisarchik A N 2019 Visual and kinesthetic modes affect motor imagery classification in untrained subjects *Sci. Rep.* **9** 9838
- [55] Pfurtscheller G and Da Silva F L 1999 Event-related EEG/MEG synchronization and desynchronization: basic principles *Clin. Neurophysiol.* **110** 1842–57
- [56] Choi J, Kim B, Huh S and Jo S 2020 Observing actions through immersive virtual reality enhances motor imagery training *IEEE Trans. Neural Syst. Rehabil. Eng.* **28** 1614–22
- [57] Chen L et al 2021 Adaptive asynchronous control system of robotic arm based on augmented reality-assisted brain-computer interface *J. Neural Eng.* **18** 066005
- [58] Tong J, Xing Z, Yue C, Dong E, Du S, Sun Z, Solés-Casals J and Caiafa C 2022 Towards improving motor imagery brain-computer interface using multimodal speech imagery *Cognit. Comput.*